



Computer Science 3B

Practical Assignment 05

Assignment date:

2026-08-20

Deadline:

2026-08-20 17h00

Marks: 120

This practical assignment must be uploaded to lms.uj.ac.za **before** 2026-08-20 17h00. Late¹ or incorrect submissions **will not be accepted**, and will therefore not be marked. You are **not allowed to collaborate** with any other student. Plagiarism is not tolerated. All submissions are tested for plagiarism.

Good coding practices include a proper coding convention and a good use of commenting. Marks will be deducted if these are not present.

The reminder page includes details for submission. Please ensure that **ALL** submissions follow the guidelines. The reminder page can be found on the last page of this assignment.

An **AI Research Consortium** is running large-scale experiments with generative models. Each experiment requires a certain amount of computing and infrastructure resources (GPUs, TPUs, CPUs, RAM and storage), and each resource has an associated per-unit cost. **Write an 80x86 assembly program²** with the following functions:

```

1// Fill an array with user input
2void inputArray(int* array, int arraySize);
3
4// Display the contents of an array
5void display(int* array, int arraySize);
6
7// Calculate the total cost given the number of resources and costs
8int totalCost(int* resources, int* costs, int arraySize);

```

In order to calculate costs the program requires two arrays: **resources**[→] which is the required amount of each resource and **costs**[→] which is the cost of each resource. Total cost can be calculated as a **dot product** of **resources**[→] and **costs**[→]!³

$$\text{resources}^{\rightarrow} \cdot \text{costs}^{\rightarrow} = \sum_{i=1}^n \text{resources}_i \times \text{costs}_i = \text{resources}_1 \times \text{costs}_1 + \dots + \text{resources}_n \times \text{costs}_n$$

Problem Decomposition:

1. Ask the user for required arrays (**resources**[→], **costs**[→]) using the **inputArray** function.
2. Display each array using the **display** function
3. Display the total cost using the **totalCost** function
4. Ask the user if they want to process another set of resources and costs, loop back to step 1 if they do.

¹ Alternate arrangements for exceptional circumstances will be posted on Moodle.

² **ESI** and **EDI** must not be used at all!

You are only allowed to create global variables for **resources**[→] and **costs**[→] as well as any string prompts. Make use of local variables as needed.

Note: The design must include a stack diagram for each function as well as **_start**.

Testing set - Use these values to test your program:

Resources (GPUs, TPUs, CPUs, RAM, Storage)	Costs per unit	Total Cost
[54, 20, 35, 54, 82]	[470,969,991, 19, 41]	83833
[23, 46, 90, 13, 32]	[738,417,410,252,619]	96140
[3, 2, 58, 55, 34]	[831,348, 14,792,932]	79249
[89, 25, 46, 79, 26]	[242,482,963,724,808]	156090
[30, 23, 44, 59, 7]	[528,697,544,241,643]	74527

Mark sheet

1. Design	[10]
2. Exit loop	[05]
3. inputArray function (entry/exit, input, loop)	[10]
4. display function (entry/exit, output, loop)	[10]
5. totalCost function (entry/exit, calculation, loop)	[15]
6. Correct display	[05]
7. Structure and layout (no temporary variables, correct data types)	[05]
8. Commenting	[05]
9. Correct execution	[55]
Total	[120]

Important:

Submissions that **do not assemble** will be capped at 40%!

Practical marks are awarded subject to the student's ability to explain the concepts and decisions made in preparing the practical assignment solution.

(Inability to explain code → inability to be given marks.)

Execution marks are awarded for a correctly functioning application and not for related code.

Reminder

Your submission must follow the naming convention below:

SURNAME_INITIALS_STUDENTNUMBER_SUBJECTCODE_YEAR_PRACTICALNUMBER

Example: Berners-Lee_TJ_209912345_CSC03B3_2026_P05

Surname	Berners-Lee	Initials	TJ	Student number	209912345
Module Code	CSC03B3	Current Year	2026	Assignment	RA

Your submission must be a **single zip (compressed) file** that includes the following:

File	Naming	Folder	Purpose
Design	STUDENTNUMBER_P05.pdf	docs	Contains your program design. All files must be in PDF format. Your details must be included at the top of any PDF files submitted ¹ .
Source	STUDENTNUMBER_P05.asm	src	Contains all relevant source code. Your details must be included at the top of the source code ⁰ .

Multiple uploads

Note that only **one** submission is marked. If you already have submitted once and want to upload a newer version then submit a newer file with the same name as the uploaded file in order to overwrite it.

¹ Failure to correctly indicate your details will result in a penalty.